

wine_analysis

April 20, 2025

```
[ ]: # Wine Quality Analysis

## Environment set up

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

## 1. Load Data

wine_red = pd.read_csv("datasets/winequality-red.csv", sep=";")
wine_white = pd.read_csv("datasets/winequality-white.csv", sep=";")
wine_red.columns = wine_red.columns.str.strip()
wine_white.columns = wine_white.columns.str.strip()

## 2. Generate plots

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.countplot(data=wine_red, x="quality", ax=axes[0])
axes[0].set_title("Red Wine Quality Distribution")

sns.countplot(data=wine_white, x="quality", ax=axes[1])
axes[1].set_title("White Wine Quality Distribution")

## 3. Show plots, and save them to figures dir as png files

plt.tight_layout()
plt.savefig('figures/wine_quality_comparison.png')
plt.show()

sns.countplot(data=wine_red, x="quality")
plt.savefig('figures/red_wine_quality.png')
plt.title("Red Wine Quality Distribution")
plt.show()
```

```
sns.countplot(data=wine_red, x="quality")  
plt.savefig('figures/white_wine_quality.png')  
plt.title("White Wine Quality Distribution")  
plt.show()
```